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Spatial Autoregressive Models for Geographically Hierarchical Data Structures

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This article discusses how standard spatial autoregressive models and their estimation can be extended to accommodate geographically hierarchical data structures. Whereas standard spatial econometric models normally operate at a single geographical scale, many geographical data sets are hierarchical in nature—for example, information about houses nested into data about the census tracts in which those houses are found. Here we outline four model specifications by combining different formulations of the spatial weight matrix W *and of ways of modeling regional effects. These are (1) groupwise* W *and fixed regional effects; (2) groupwise* W *and random regional effects; (3) proximity-based* W *and fixed regional effects; and (4) proximity-based* W *and random regional effects. We discuss each of these model specifications and their associated estimation methods, giving particular attention to the fourth. We describe this as a hierarchical spatial autoregressive model. We view it as having the most potential to extend spatial econometrics to accommodate geographically hierarchical data structures and as offering the greatest coming together of spatial econometric and multilevel modeling approaches. Subsequently, we provide Bayesian Markov Chain Monte Carlo algorithms for implementing the model. We demonstrate its application using a two-level land price data set where land parcels nest into districts in Beijing, China, finding significant spatial dependence at both the land parcel level and the district level.*

Introduction

Many geographical data sets have hierarchical structures—for example, houses nested into districts into regions in an urban housing market (Jones 1991). Using the language of the multilevel modeling literature (Goldstein 2003), the finer spatial scale at which outcomes of interest are measured is termed the lower level, and the higher levels are the more aggregate spatial scales such as regions. The assumption of multilevel modeling is that there is both difference between objects at the higher level and some level of group dependence within those objects. In other words, difference between regions and correlations within regions are expected:

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the outcomes of lower level units in the same region tend to be correlated because they are impacted by the same effects. This can be described as vertical group dependence.

However, horizontal dependence cannot be modeled using classic multilevel approaches. This is the sort of spatial dependence associated with spatial econometric modeling of single-level spatial data sets; it arises from the interactions or spillovers among spatial entities due to their geographic proximity. When working with geographically hierarchical data sets, we can expect (or at least allow for) both types of dependence: the vertical and the horizontal. The first, a top-down group dependence, is due to regional (or contextual) effects; the second to the traditional spatial dependence concern of spatial econometrics. Essentially, the distinction we are making is between spatial interaction (at lower levels) and spatial heterogeneity (at regional levels). Defined as such, both types of dependency are clearly spatial. However, the phrase group dependency acknowledges that the allocation of individuals to regions need not be geographical (hierarchical models need not be modeling a geographical hierarchy).

Conceptually, in this article, we distinguish these two types of dependency by providing a general methodological framework for simultaneously modeling the spatial interaction and regional effects. The merits associated with this methodology are several. Generally, it will provide more efficient and accurate model estimation for regression coefficients by recognizing the hierarchical data structure. The regional effects estimated by the model can be mapped to explicitly investigate the spatial patterns in the outcomes of interest at the regional level. In addition, it provides more accurate estimates of the intensity of spatial interaction at the lower level by separating measurement from the regional effects with which it may be confounded.

We build the methodological framework based on the standard spatial econometric methods, especially the spatial simultaneous autoregressive model (Cliff and Ord 1981; Anselin 1988; Griffith 1988; Haining 2003; LeSage and Pace 2009). Within spatial econometrics, there has been interest in the use of hierarchical data sets where regional variables are used to control possible regional effects when modeling outcomes observed at a subregional scale. For example, when modeling house prices, regional-level variables such as crime rates, school quality, or socioeconomic description are incorporated into the model using data for census tracts or districts (see, e.g., Bowen, Mikelbank, and Prestegaard 2001; Baumont 2009). The logic is that the spatial interaction effect indicated by a significant autoregressive parameter (in a lagged y model: LeSage and Pace 2009, pp. 16–20) might merely be a statistical artifact created by ignoring regional effects. Unfortunately, simply adding regional variables is not sufficient to model the outcome of interest appropriately. Treating a regional covariate as a repeated measurement, whereby all individuals within the same region are assigned identical values of the regional variable, will lead to inefficient model estimation and incorrect statistical inference (Goldstein 2003).

Given the prevalence of geographically hierarchical data sets, it is surprising that studies on modifying standard spatial autoregressive (SAR) models to better accommodate a geographically hierarchical structure remain rare (exceptions being Smith and LeSage 2004; Corrado and Fingleton 2012; LeSage and Llano 2013). In the context of modeling 1996 presidential election results for U.S. counties, Smith and LeSage (2004) build a spatial probit model in which election outcomes of each county are not independent of each other. The dependence is due to these counties being taken to be nested into U.S. states. It is assumed that counties within the same state share the same state-level (regional) effect. Moreover, these regional effects themselves are spatially dependent at the state level. There are spatial patterns to voting at the state level, and the

interaction between these state-level voting behaviors is estimated with a spatial interaction term. What is not considered is the equivalent interaction at a county level. The assumption that the lower level interaction can be discarded or is subsumed by the higher level interaction is not reasonable in many situations (Case 1992; McMillen 1992; LeSage 2000; LeSage et al. 2011). Corrado and Fingleton (2012) propose an approach to estimate spatial dependence among lower level units (specified in a groupwise way, see below) by finding valid instrumental variables for the endogenous lagged outcome variable. The concern here is that their groupwise specification of lower level spatial dependence violates the notion of distance decay, the core concept in spatial analysis (Getis and Ord 1992; Haining, Kerry, and Oliver 2010; Anselin and Arribas-Bel 2013). Moreover, the regional effects in their framework are assumed to be independent rather than dependent.

What we are proposing is similar to the spatial misalignment perspective on hierarchical structures of spatial data that is found in the spatial statistics literature. The problem of spatial misalignment, or a change of support problem, refers to analysis or inference on a variable at one spatial scale based on data at another spatial hierarchy, or on relationships between spatial variables at different scales (Gotway and Young 2002; Banerjee, Carlin, and Gelfand 2004, p. 175). For example, in an analysis of the impact of air pollution on health outcomes, the air pollution data are from a number of monitoring locations scattered across the study area whereas the health outcomes are only available at the scale of census tracts. The solution is in two steps (Agarwal, Gelfand, and Silander 2002; Zhu, Carlin, and Gelfand 2003). The first step is to "realign" variables to the same spatial scale by interpolating census tract-level air pollution from point-level information (using methodologies described by Mugglin and Carlin 1998; Gelfand, Zhu, and Carlin 2001). The next step is to relate the health outcomes to the newly realigned census tract air pollution variable.

In particular, Banerjee, Carlin, and Gelfand (2004, pp. 206–212) demonstrate a two-level spatially misaligned regression model where the interest is to explain the relationship between land uses and population at the pixel level but with a key problem being that the population variable is only observed at the town level. They have constructed a full Bayesian framework for estimating such types of spatial multilevel models to capture the impact of the latent pixel-level population on land uses by conditioning on the higher level town population. The similarities between their framework and our approach are in modeling a spatially hierarchical data structure and examining the impacts of covariates at both the lower and higher levels on outcomes measured at the lower level. However, several differences can also be identified. First, their higher level variable plays an important and explicit role in inferring the relationship between the corresponding unobserved variable at the lower level and the outcome variable. By contrast, in our approach, the higher level covariate effects are directly interpreted as the observed contextual effects on outcomes measured at the lower level in a way that is similar with a typical regression model. Another important difference is that spatial interaction effects at the lower level are not modeled in their framework, which is a major concern in this article.

Proximity-based and groupwise spatial weight matrices and hierarchical data structures

In spatial econometrics, the spatial connectivity matrix, W, specifies the interaction structure among the modeled entities. It defines the connection between those entities that will be examined for spatial dependence. Most often the strength of spatial dependency is assumed to be proximity based, defined by geographic distance, nearest neighbors, or by contiguity (see detailed

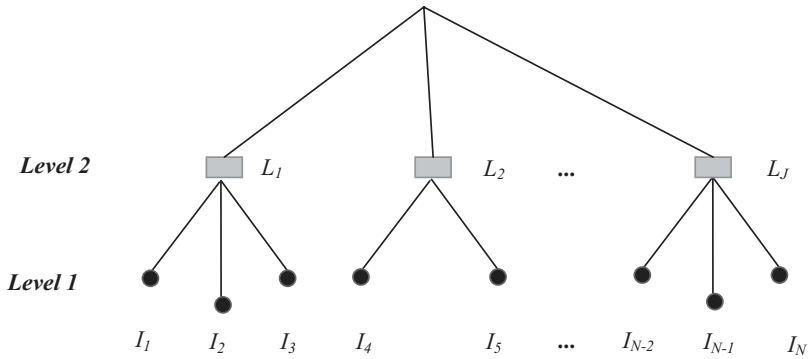

Figure 1. A two-level hierarchical data structure.

discussions in Cliff and Ord 1981; Griffith 1988, 2003; Bavaud 1998). However, W could also define purely groupwise effects, regarding entities within the same region as neighbors of each other and leaving entities between regions as unconnected. This groupwise W has been discussed extensively in the social interaction literature where the primary interest is to estimate the endogenous group effects or peer effects on an individual's behavior (Manski 1993; Durlauf and Ioannides 2010; Lin 2010). The statistical properties of estimators using this groupwise W have been investigated by Lee (2007; see also Kelejian and Prucha 2002; Lin 2010; Anselin and Arribas-Bel 2013). The essential departure of the groupwise spatial dependence from the proximity-based dependence is that the former conceptualizes spatial dependence as discontinuous, bounded by regional boundaries. Consequently, two lower level units located either side of a boundary are assumed independent even though they are in close proximity.

Given the groupwise spatial dependence specification, W is reduced to a block diagonal matrix. Following the notation in Corrado and Fingleton (2012), for J groups or regions, W is

$$W = \begin{bmatrix} W_1 & 0 & \cdots & 0 \\ 0 & W_2 & \cdots & 0 \\ \cdots & \cdots & \cdots & \cdots \\ 0 & 0 & \cdots & W_J \end{bmatrix} \tag{1}$$

where $W_j = 1/nj \times (\mathbf{1}_{nj}\mathbf{1}'_{nj})$, nj is the number of lower level units in group j, $\boldsymbol{1}_{nj}$ is a $nj \times 1$ column vector of ones.

In this article, we focus on extending SAR models for a two-level hierarchical data structure. Slightly adapting the diagrammatic description in Corrado and Fingleton (2012, p. 227, fig. 4), a typical two-level geographically hierarchical data structure can be represented as in Fig. 1.

Here L_j ($j = 1, 2 \ldots J$) denotes regions, as discussed above, and I_i ($i = 1, 2 \ldots N$) indicates lower level units. Under this data structure, lower level units within the same region are correlated as they are exposed to the same regional effects.

SAR models for hierarchical data structures

Following LeSage and Pace (2009), the standard SAR model (the spatial lag model in Anselin 1988) can be specified as

$$Y = \rho WY + X\beta + \varepsilon \quad \varepsilon \sim N\left(0, I_N \sigma^2\right) \tag{2}$$

where Y is an N by 1 column vector of outcome variable values; X is an N by K matrix denoting the independent variables; ρ is the SAR parameter indicating the strength of spatial interaction; β is a K by 1 vector of regression coefficients to estimate; and ε is an independent, normally distributed error term.

To incorporate the regional effects, the SAR model can be modified as

$$Y = \rho WY + X\beta + Z\gamma + \Delta\boldsymbol{\theta} + \varepsilon \tag{3}$$

where Δ is a N by J block diagonal design matrix with a structure in the form of equation (1); Z is an N by P matrix consisting of higher level variables, and γ is the corresponding P by 1 coefficient vector to estimate. We note that the value for each regional variable is the same for all lower level units belonging to the same region; therefore, Z could be replaced by ΔZ^* where Z^* is a J by P matrix.

The J by 1 column vector θ, θ $[\theta_1, \theta_2, \ldots, \theta_J]$ denotes the regional effects that could be treated as either fixed or random. It is useful to note that, in the case of fixed group effects, Δ should be an N by $(J - 1)$ block diagonal design matrix as the intercept term is in the X. For simplicity, we assume that $\varepsilon \sim N\left(0, I_N \sigma_e^2\right)$. When θ is assumed as random effects, θ has a normal distribution with mean equal to 0 and covariance $\sigma_u^2 \Omega_u$, $\theta \sim N\left(0, \sigma_u^2 \Omega_u\right)$. If θ is independent, Ω_u reduces to a diagonal matrix I_J; otherwise, Ω_u represents the correlation structure for θ.

Given the two specifications of W, proximity-based W and groupwise-based W, and also noting that the regional effects can be treated as either fixed or random, then equation (3) can be conceptualized as four different model specifications, as shown in Fig. 2. We now consider each of these models in turn.

Model (a): Groupwise *W* and fixed-effects θ

The first of the models shown in Fig. 2 is groupwise spatial dependence with fixed regional effects. For expositional clarity, we re-express equation (3) as

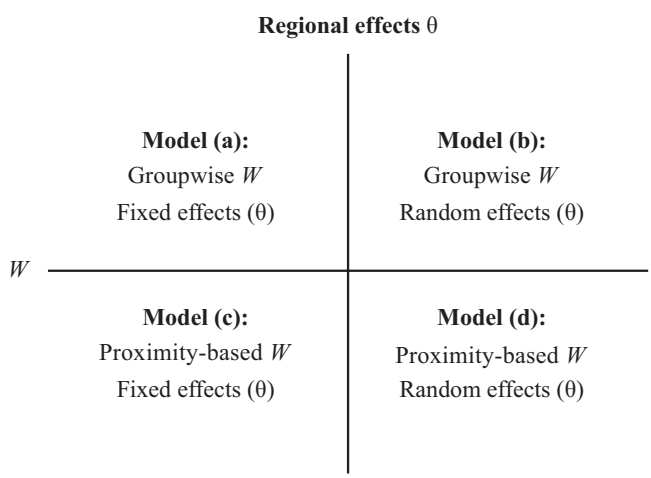

Figure 2. A typology of four model specifications from combining different forms of ***W*** and θ.

$$y_{ij} = \rho W_i Y + x_{ij}\beta + \theta_j + \varepsilon_{ij} \tag{4}$$

where i and j indicate the lower level and regional level, respectively. We note that regional variables cannot be included in the model specification with fixed regional effects. $W_i = [w_{i1}, w_{i2}, \ldots, w_{iN}]$ denotes the ith row of the groupwise W and θ_j is the jth regional effect. Under the assumption of groupwise spatial dependence, $W_i Y$ is equal to the mean of the jth region, $\bar{y}_J$. Then equation (4) is equivalent to

$$y_{ij} = \rho\bar{y}_J + x_{ij}\beta + \theta_j + \varepsilon_{ij} \tag{5}$$

Because $\bar{y}_J$ varies only between regions, the SAR parameter ρ cannot be identified in this model because of the fixed regional effects (ρ is also known as the endogenous group effect in the situation of groupwise W; see Corrado and Fingleton 2012). If we assume that lower level units cannot influence themselves, the structure of the groupwise W defined above can be changed slightly by re-assigning the diagonal elements to 0 s. Consequently, $W_i Y$ will not be exactly equal to $\bar{y}_J$, as W_{ii} is 0. Even so, and as Anselin and Arribas-Bel (2013), based on Kelejian and Prucha (2002) show, $W_i Y$ remains very close to a constant for units within the same region. Therefore, ρ can be very weakly identified at best. If the spatial dependence effect exists in the form of groupwise dependence, then $\rho\bar{y}_J$ can be dropped from the model and the fixed-effect models will then fully control this form of spatial dependence and provide consistent estimates of β (Anselin and Arribas-Bel 2013).

Model (b): Groupwise *W* and random effects θ

This is the second of the models in Fig. 2. Under the assumption of random regional effects θ and further assumption of independence between θ and ε, the question of interest in the social interaction literature is whether the model parameters, especially ρ, are identifiable.

As a simple illustration, let $x_{ij} = [1, x_{ij}, z_j]$ and $\beta = [\beta_0, \beta_1, \beta_2]$. Given this, equation (5) can be rewritten as

$$y_{ij} = \rho\bar{y}_J + \beta_0 + x_{ij}\beta_1 + z_j\beta_2 + \theta_j + \varepsilon_{ij} \tag{6}$$

where θ is distributed as $N\left(0, I_J\sigma_u^2\right)$, z_j is a regional-level variable, and β_2 represents the contextual effect.

Following Corrado and Fingleton (2012, p. 232), after taking group/regional means of both sides of equation (6), and rearranging, we obtain

$$\bar{y}_J = (1-\rho)^{-1}\beta_0 + (1-\rho)^{-1}\beta_1\bar{x}_j + (1-\rho)^{-1}\beta_2 z_j + \theta'_j \tag{7}$$

where $\rho < 1$, which is true when W is normalized either through row normalization or adjusted by the maximum eigenvalue of W (for detailed discussions, see Cliff and Ord 1981; Griffith 1988; Haining 2003; LeSage and Pace 2009), and where $\theta'_j = (1-\rho)^{-1}\left(\bar{\varepsilon}_J + \theta_j\right)$.

Then we take equation (7) into equation (6) and center x_{ij} by the group means, from which equation (8) is obtained:

$$y_{ij} = (1-\rho)^{-1}\beta_0 + (1-\rho)^{-1}\beta_1\bar{x}_J + \beta_1\left(x_{ij} - \bar{x}_J\right) + (1-\rho)^{-1}\beta_2 z_j + \theta''_j + \varepsilon_{ij} \tag{8}$$

where $\theta''_j = \rho\theta'_j + \theta_j$. As illustrated by Corrado and Fingleton (2012), under the assumption of the equivalence of z_j and $\bar{x}_J$, the classic Manski's reflection problem will emerge: the endogenous

group effects ρ and contextual effects β_2 are not identifiable simultaneously (Manski 1993). They further show that if both these parameters need to be identified, the dependence between groups like the out-of-group z_j and $\bar{x}_J$ should also be included in the model specification.

Model (c): Proximity-based *W* and fixed-effects θ

For the third of the model types, we can re-arrange equation (3) as

$$Y = \rho WY + X\beta + \Delta\boldsymbol{\theta} + \varepsilon. \tag{9}$$

For the estimation of this model, we first eliminate the fixed regional effects θ using a de-group-mean transformation (see, e.g., Lee 2007; Lin 2010) by multiplying equation (9) by a block diagonal transformation matrix Q. Q = diagonal $\{Q_1, Q_2, \ldots, Q_J\}$ where the jth block is Q_j:

$$Q_j = \boldsymbol{I}_{nj} - nj^{-1} \times \mathbf{1}_{nj}\mathbf{1}'_{nj} \tag{10}$$

in which nj and $\mathbf{1}_{nj}$ have the same definition as above, and $\boldsymbol{I}_{nj}$ is an identity matrix with order of nj. After this, equation (9) is changed as

$$Y^* = \rho WY^* + X^*\beta + \varepsilon^* \tag{11}$$

where $Y^* = QY$, $X^* = QX$ and $\varepsilon^* = Q\varepsilon$.

On first inspection, it seems that equation (11) is a SAR model as in equation (2), and that standard estimation methods like maximum likelihood estimation (MLE) could therefore be employed directly. However, as pointed out by Lee (2007) and Lin (2010), the transformed error terms, ε*, are not independent and identically distributed as is assumed of ε in equation (2). In contrast, they are heteroskedastic and groupwise correlated. Therefore, in order to employ the standard estimation method for SAR models, those authors propose further transformation of equation (11) using eigenvalue–eigenvectors decomposition techniques to ensure the error terms in the equation are independent and homoscedastic.

Model (d): Proximity-based *W* and random effects θ

Finally, in the case of model (d), the model is specified as

$$\begin{aligned} &Y = \rho WY + X\beta + Z\gamma + \Delta\boldsymbol{\theta} + \varepsilon \\ &\operatorname{cov}(\varepsilon) = I_N\sigma_e^2; \operatorname{cov}(\boldsymbol{\theta}) = \sigma_u^2\Omega_u; \operatorname{cov}(\varepsilon, \boldsymbol{\theta}) = 0 \end{aligned} \tag{12}$$

Instead of assuming independent random regional effects θ, as in the standard multilevel modeling literature (Jones 1991; Goldstein 2003), we relax this restriction by allowing for dependent θ. In the context that spatial regions are also geographically contiguous, it is expected that the effect of region j, θ_j may be similar to that of its neighboring regions. Following Smith and LeSage (2004) and in line with the spatial econometrics literature (e.g., Cliff and Ord 1981; Griffith 1988, 2003; Anselin 1988; LeSage and Pace 2009), we postulate that the random regional effects follow a spatial simultaneous autoregressive specification:

$$\boldsymbol{\theta} = \lambda M\boldsymbol{\theta} + u, \quad u \sim N\left(0, I_J\sigma_u^2\right) \tag{13}$$

where M is a normalized spatial weight matrix at the region level (e.g., defined by geographic distance, nearest neighbors, or by contiguity), and λ measures the extent to which these regional

effects are spatially interacted. The residuals u are assumed to be independent normal variates. Solving equation (13), we get

$$\boldsymbol{\theta} = (I_J - \lambda M)^{-1} u = B^{-1} u \tag{14}$$

where $B = I_J - \lambda M$. Therefore, the covariance matrix for $\boldsymbol{\theta}$, $\operatorname{cov}(\boldsymbol{\theta}) = \sigma_u^2 \Omega_u = \sigma_u^2 (B'B)^{-1}$ given the non-singular B. The distribution of θ is multivariate normal with means 0 s and covariance $\sigma_u^2 \Omega_u$, thus $\boldsymbol{\theta} \sim N\left(0, \sigma_u^2 (B'B)^{-1}\right)$.

Combining equation (12) and (13), we get our complete specification for model (d):

$$Y = \rho WY + X\beta + Z\gamma + \Delta\boldsymbol{\theta} + \varepsilon; \quad \boldsymbol{\theta} = \lambda M\boldsymbol{\theta} + u \tag{15}$$

In the subsequent analysis, we term this specification a hierarchical SAR model (HSAR) in the sense that it extends the standard SAR model to accommodate hierarchical data structures. Recalling that the motivation of this study is to distinguish the horizontal spatial dependence and vertical group dependence arising from (dependent) regional effects θ, it is helpful to write the covariance matrix of Y, which is

$$\operatorname{cov}(Y) = A^{-1} \operatorname{cov}(\Delta\boldsymbol{\theta} + \varepsilon)\left(A^{-1}\right)^{\mathrm{T}} = \sigma_e^2 \left(A^{\mathrm{T}} A\right)^{-1} + \sigma_u^2 \left(A^{-1}\Delta\right)\Omega_u \left(A^{-1}\Delta\right)^{\mathrm{T}} \tag{16}$$

where $A = I_N - \rho W$. The first part of the covariance matrix of Y is due to the simultaneous autoregressive process of Y, that is, the horizontal spatial interactions among lower level units. The second component refers to the influence of group dependence. To see the group dependence more clearly, imagine an extremely simple case where there is no spatial dependence ($\rho = 0$) and independent group effects θ ($\lambda = 0$). The covariance matrix for Y is then $\left(\sigma_e^2 I_N + \sigma_u^2 \Delta\Delta^{\mathrm{T}}\right)$ and the group dependence is measured by $\left(\sigma_u^2 / \left(\sigma_e^2 + \sigma_u^2\right)\right)$.

As is well established by LeSage and Pace (2009), because of the spatial interaction effect (through ρ), the marginal effect of a covariate (like $\partial y/\partial X_k$) normally does not equal the estimated coefficient β_k in a SAR model. The same situation is true for the HSAR. For deriving marginal effects for each covariate, we rewrite equation (15) as one equation by replacing θ with equation (14) and Z with ΔZ^*:

$$Y = (I_N - \rho W)^{-1} \left(X\beta + \Delta Z^* \gamma + \Delta B^{-1} u + \varepsilon\right). \tag{17}$$

The marginal effects for a lower level variable x_k in HSAR is

$$\begin{aligned} \partial y / \partial x_k &= (I_N - \rho W)^{-1} I_N \beta_k = S_k(W) I_N \beta_k \\ S_k(W) &= (I_N - \rho W)^{-1} = \left(I_N + \rho W + \rho^2 W^2 + \ldots\right) \end{aligned} \tag{18}$$

which is exactly the same as in the SAR model (see LeSage and Pace 2009, p. 38) because the regional effects are irrelevant to X. The total impact on each lower level unit of a one unit increase associated with x_k is the sum of each row of $S_k(W) I_N \beta_k$ through multiplying equation (18) by N by 1 vector of ones, l_N:

$$(\partial y / \partial x_k)_{N \times 1} = S_k(W) I_N \beta_k l_N = \left(1 + \rho + \rho^2 + \ldots\right) \beta_k l_N = \beta_k / (1 - \rho) l_N \tag{19}$$

which is N by 1 vector of $\beta_k / (1 - \rho)$ and thus the average total impacts is also $\beta_k / (1 - \rho)$. In other words, a one unit change of x_k induces the same amount of increase/decrease in y for each

observation. However, this situation could be easily modified, for example, by including the spatially lagged X, WX. Then the total effects of a unit change of x_k for each observation will vary (LeSage and Pace 2009; Elhorst 2010). The direct impact on each observation of a unit increase associated with x_k is the diagonal elements of $S_k(W)I_N\beta_k$, which is observation specific. The average direct impact is calculated by $trace(S_k(W)I_N\beta_k)/\mathrm{N}$. The indirect impact is obtained by subtracting the direct impact from the total impacts.

Similarly, the marginal effects for a regional variable z_p are

$$\partial \mathrm{y}/\partial z_p = (I_N - \rho W)^{-1}\Delta I_J \gamma_p = I_N \Delta\gamma_p + \rho W \Delta\gamma_p + \rho^2 W^2 \Delta\gamma_p + \ldots \tag{20}$$

The first term $I_N\Delta\gamma_p$ implies that the impact of a unit change in z_p is shifted to all observations with magnitude of γ_p. The second term $\rho \mathrm{W}\Delta\gamma_p$ reflects the further influence of the N by 1 vector $\gamma_p l_N$ through the first-order spatial weight matrix W discounted by ρ. These impacts repeat for higher order spatial weight matrices (like W^2, W^3, . . .) attenuated by corresponding powers of ρ (e.g., ρ^2, ρ^3, . . .).[1] To see the similarity with the impacts of lower level variables as to the (average) total impacts, we multiply equation (20) by a J by 1 vector of ones, l_J:

$$\begin{aligned}(\partial y/\partial z_p)_{N\times 1} &= I_N \Delta l_J \gamma_p + \rho W \Delta l_J \gamma_p + \rho^2 W^2 \Delta l_J \gamma_p + \ldots \\ &= (1+\rho+\rho^2+\ldots)\gamma_p l_N = \gamma_p/(1-\rho) l_N\end{aligned} \tag{21}$$

because $\Delta l_J = l_N$ and W is row normalized. Here, we obtain the same structure for the (average) total impacts associated with a unit change in z_p as that for x_k. To modify this, we could add spatially lagged Z to produce varying total effects for each observation. The direct impacts on each individual of a unit change in z_p is the elements in $S_p(W)\Delta I_J\gamma_p$ whose location corresponds to that of 1 s in the random effect design matrix Δ, or the diagonal elements of $S_p(W)\Delta I_J\gamma_p\Delta^{\mathrm{T}}$. Similarly, the average direct impacts is $trace(S_p(W)\Delta I_J\gamma_p\Delta^{\mathrm{T}})/N$ and the indirect impact is the total impact subtracting the direct impact.

Estimating the HSAR model

As indicated by the references in the preceding discussion, the estimation of model (a), (b) and (c) has been investigated. However, to the best of our knowledge, the HSAR model has not been discussed before. Therefore, in this section, we provide Bayesian Markov Chain Monte Carlo (MCMC) algorithms for implementing it. The application of Bayesian estimation methods in spatial econometrics has been discussed by many researchers (inter alia, Hepple 1995a; LeSage 1997; LeSage and Pace 2009). The Bayesian MCMC method is preferred over MLE in the spatial econometrics literatures for implementing complex models such as those with a complex variance–covariance structure or with limited dependent variables (LeSage 1997, 2000), for spatial panel data models with random effects (Parent and LeSage 2012), and for undertaking model comparisons (Hepple 1995b; LeSage and Parent 2007). The estimation of HSAR is achieved via MCMC methods that draw samples sequentially from the conditional posterior distributions for each model unknown parameter (Gelfand and Smith 1990; Smith and LeSage 2004). To implement HSAR, we need to specify the prior distributions for each parameter and derive their conditional distributions.

Prior specifications for HSAR

In the Bayesian paradigm, the basic principle is that the posterior distribution of θ* is proportional to the product of the data and prior distributions, as denoted by

$$P\left(\boldsymbol{\theta}^* \mid \text{Data}\right) \propto P\left(\text{Data} \mid \boldsymbol{\theta}^*\right) \times P\left(\boldsymbol{\theta}^*\right). \tag{22}$$

We note that as there is no difference with respect to estimating regression coefficient sets, β and γ, we let $X = [X\ Z]$ and $\beta = [\beta, \gamma]$ for notational convenience. Therefore, in the HSAR, $\boldsymbol{\theta}^* = \left\{\rho, \lambda, \boldsymbol{\beta}, \boldsymbol{\theta}, \sigma_e^2, \sigma_u^2\right\}$, Data = $\{Y, X, W, M\}$. The prior distribution P (θ*) reflects the prior knowledge and uncertainty on θ*, and the posterior distribution P (θ* | Data) reflects the knowledge on θ* by updating the prior knowledge with sample data information (LeSage and Pace 2009, p. 125). The inferences are based on the posterior distributions of model parameters.

In this article, we utilize a k-dimensional multivariate normal prior for β, with mean M_0 and variance matrix T_0; therefore, $P(\beta) \sim N(M_0, T_0)$. We assign a uniform prior distribution for ρ over a feasible range $[1/v_{min}, 1/v_{max}]$, where v_{min} and v_{max} represent the minimum and maximum eigenvalues of the spatial connectivity matrix W. When a row normalization procedure is applied to W, $v_{max} = 1$, and then the range becomes $[1/v_{min}, 1]$ (LeSage and Pace 2009). Similarly, we assume a uniform prior distribution for λ over $[1/v^*_{min}, 1]$ where v^*_{min} is the minimum eigenvalue of the higher level spatial weight matrix M. $P\left(\sigma_e^2\right)$ and $P\left(\sigma_u^2\right)$ are assigned an inverse gamma distribution, $P\left(\sigma_e^2\right) \sim IG\left(c_0, d_0\right)$ and $P\left(\sigma_u^2\right) \sim IG\left(a_0, b_0\right)$. Following Gelman et al. (2003), the inverse gamma distribution (non-normalized) with a and β being the shape parameter and scale parameter is parameterized as

$$p(x) \propto x^{-\alpha-1} exp(-\beta/x) \tag{23}$$

In order to derive the posterior distribution for unknown parameters, we need first to write the likelihood function that indicates P (Data | θ*). The likelihood function for the HSAR model can be expressed as

$$L\left(Y|\rho, \lambda, \boldsymbol{\beta}, \boldsymbol{\theta}, \sigma_u^2, \sigma_e^2\right) = \left(2\pi\sigma_e^2\right)^{-N/2} |A| \exp\left\{-\left(2\sigma_e^2\right)^{-1}(AY - X\boldsymbol{\beta} - \Delta\boldsymbol{\theta})'(AY - X\boldsymbol{\beta} - \Delta\boldsymbol{\theta})\right\} \tag{24}$$

where $A = \boldsymbol{I}_N - \rho W$, as in equation (16) and $|A|$ denotes the determinant of A. Therefore, the full posterior distribution for $\boldsymbol{\theta}^* = \left\{\rho, \lambda, \boldsymbol{\beta}, \boldsymbol{\theta}, \sigma_e^2, \sigma_u^2\right\}$ is

$$\begin{aligned} P\left(\rho, \lambda, \boldsymbol{\beta}, \boldsymbol{\theta}, \sigma_e^2, \sigma_u^2 \mid Y\right) \propto L\left(Y \mid \rho, \lambda, \boldsymbol{\beta}, \boldsymbol{\theta}, \sigma_e^2, \sigma_u^2\right) \times P\left(\boldsymbol{\beta}\right) \times P\left(\rho\right) \times P\left(\lambda\right) \times P\left(\boldsymbol{\theta} \mid \lambda, \sigma_u^2\right) \\ \times P\left(\sigma_u^2\right) \times P\left(\sigma_e^2\right) \end{aligned} \tag{25}$$

Deriving the conditional posterior distribution for each parameter

The conditional posterior distribution for β

Let the posterior distribution for β, $P\left(\boldsymbol{\beta} \mid Y, \rho, \lambda, \boldsymbol{\theta}, \sigma_e^2, \sigma_u^2\right) \sim N\left(M_\beta, \Sigma_\beta\right)$. According to equation (25), the posterior distribution for β is

$$\begin{aligned} P\left(\boldsymbol{\beta} \mid Y, \rho, \lambda, \boldsymbol{\theta}, \sigma_e^2, \sigma_u^2\right) &\propto L\left(Y \mid \rho, \lambda, \boldsymbol{\beta}, \boldsymbol{\theta}, \sigma_e^2, \sigma_u^2\right) \times P(\boldsymbol{\beta}) \\ &\propto \exp\left\{-\left(2\sigma_e^2\right)^{-1}(AY - X\boldsymbol{\beta} - \Delta\boldsymbol{\theta})'(AY - X\boldsymbol{\beta} - \Delta\boldsymbol{\theta})\right\} \times \exp\left\{-\frac{1}{2}(\boldsymbol{\beta} - M_0)'T_0^{-1}(\boldsymbol{\beta} - M_0)\right\} \\ &\propto \exp\left\{-\frac{1}{2} \times \boldsymbol{\beta}'\left[\left(\sigma_e^2\right)^{-1} X'X + T_0^{-1}\right]\boldsymbol{\beta} + \left[\left(\sigma_e^2\right)^{-1}(AY - \Delta\boldsymbol{\theta})'X + T_0^{-1}M_0\right]\boldsymbol{\beta} + con\right\} \end{aligned} \tag{26}$$

Therefore, comparing this with a standard multivariate normal density, we obtain

$$\Sigma_{\boldsymbol{\beta}} = \left[\left(\sigma_e^2\right)^{-1} X'X + T_0^{-1}\right]^{-1}; M_{\boldsymbol{\beta}} = \Sigma_{\boldsymbol{\beta}}\left[\left(\sigma_e^2\right)^{-1} X'(AY - \Delta\boldsymbol{\theta}) + T_0^{-1}M_{\mathbf{0}}\right] \tag{27}$$

The conditional posterior distribution for $\left\{\boldsymbol{\theta}, \sigma_u^2, \sigma_e^2\right\}$

Following similar procedures to those used in deriving the posterior distribution for β, we show the posterior distribution for θ is also a multivariate normal N (M_θ, Σ_θ) where

$$\Sigma_{\boldsymbol{\theta}} = \left[\left(\sigma_e^2\right)^{-1} \Delta'\Delta + \left(\sigma_u^2\right)^{-1} B'B\right]^{-1}; M_{\boldsymbol{\theta}} = \Sigma_{\boldsymbol{\theta}}\left[\left(\sigma_e^2\right)^{-1} \Delta'(AY - X\boldsymbol{\beta})\right] \tag{28}$$

The full conditions for σ_u^2 and σ_e^2 are IV (a_u, b_u) and IV (c_e, d_e) where

$$a_u = J/2 + a_0; b_u = \boldsymbol{\theta}'B'B\boldsymbol{\theta}/2 + b_0 \tag{29}$$

$$c_e = N/2 + c_0; d_e = \frac{1}{2} \times (AY - X\boldsymbol{\beta} - \Delta\boldsymbol{\theta})'(AY - X\boldsymbol{\beta} - \Delta\boldsymbol{\theta}) + d_0 \tag{30}$$

The conditional posterior distribution for ρ

To this point we have derived the posterior distributions for $\left\{\boldsymbol{\beta}, \boldsymbol{\theta}, \sigma_e^2, \sigma_u^2\right\}$, which are all recognizable standard distributions. Gibbs sampler is therefore employed to draw samples (Gelfand and Smith 1990; Gelman et al. 2003). The posterior distribution for ρ is $P\left(\rho \mid Y, \lambda, \boldsymbol{\beta}, \sigma_e^2, \sigma_u^2, \boldsymbol{\theta}\right) \propto L\left(Y \mid \rho, \lambda, \boldsymbol{\beta}, \boldsymbol{\theta}, \sigma_e^2, \sigma_u^2\right) \times P(\rho)$

$$\propto |\boldsymbol{I}_N - \rho W| \times \exp\left\{-\left(2\sigma_e^2\right)^{-1}(AY - X\boldsymbol{\beta} - \Delta\boldsymbol{\theta})'(AY - X\boldsymbol{\beta} - \Delta\boldsymbol{\theta})\right\} \tag{31}$$

Unlike $\left\{\boldsymbol{\beta}, \boldsymbol{\theta}, \sigma_e^2, \sigma_u^2\right\}$, this conditional distribution does not fit a standard recognizable density distribution. Therefore, an alternative MCMC method, the Metropolis–Hastings (M–H) sampling strategy could be employed for updating ρ. As an alternative to the M–H sampling strategy, Smith and LeSage (2004) propose an inversion sampling procedure to update ρ and they find this method is more efficient than the M–H algorithm. The logic is that, in one MCMC loop, after updating $\left[\lambda, \boldsymbol{\beta}, \boldsymbol{\theta}, \sigma_e^2, \sigma_u^2\right]$, we conduct a numeric integration for ρ, calculate the normalized constant for the distribution of ρ, and draw from the cumulative distribution of ρ.

We employ the inverse sampling strategy in the subsequent analysis. Following Pace and Barry (1997), and given updated $\left[\lambda, \boldsymbol{\beta}, \boldsymbol{\theta}, \sigma_e^2, \sigma_u^2\right]$ in MCMC loop k, equation (31) is a function of ρ, and the non-normalized density distribution indicated by f (ρ). Log f (ρ) can be expressed as

$$\begin{aligned} &\text{Log } f(\rho) = \log|\boldsymbol{I}_N - \rho W| + S(\rho)'S(\rho)/2\sigma_e^2 + m; S(\rho) = e_0 - \rho e_d - e_u \\ &e_0 = Y - X\beta_0; e_d = WY - X\beta_d; e_u = \Delta\boldsymbol{\theta} - X\beta_u \\ &\beta_0 = (X'X)^{-1} X'Y; \beta_d = (X'X)^{-1} X'WY; \beta_u = (X'X)^{-1} X'\Delta\boldsymbol{\theta} \end{aligned} \tag{32}$$

where m is a constant. Therefore, in each loop, we can numerically integrate $\log f(\rho)$ over $(1/v_{min}, 1)$ and calculate the cumulative distribution of ρ; then the inverse sampling strategy can be employed.

The conditional posterior distribution for λ

The conditional posterior distribution for λ,

$$P\left(\lambda \mid Y, \rho, \boldsymbol{\beta}, \sigma_e^2, \sigma_u^2, \boldsymbol{\theta}\right) \propto P\left(\theta \mid \lambda, \sigma_u^2\right) \times P(\lambda) \propto \left|\boldsymbol{I}_J - \lambda M\right| \times \exp\left\{-\left(2\sigma_u^2\right)^{-1} \boldsymbol{\theta}' B' B \boldsymbol{\theta}\right\} \tag{33}$$

is not a recognizable density function as neither is $P\left(\rho \mid Y, \lambda, \boldsymbol{\beta}, \sigma_e^2, \sigma_u^2, \boldsymbol{\theta}\right)$. We also employ the above-mentioned inverse sampling strategy to update λ. Let $f(\lambda)$ indicate the non-normalized probability density, therefore, $\log f(\lambda)$ is

$$\begin{aligned} \text{Log} f(\lambda) &= \log\left|\boldsymbol{I}_J - \lambda M\right| + \boldsymbol{\theta}' B' B \boldsymbol{\theta} / 2\sigma_u^2 \\ &= \log\left|\boldsymbol{I}_J - \lambda M\right| + \left(\boldsymbol{\theta}'\boldsymbol{\theta} - 2\lambda \boldsymbol{\theta}' M \boldsymbol{\theta} + \lambda^2 \boldsymbol{\theta}' M' M \boldsymbol{\theta}\right) / 2\sigma_u^2. \end{aligned} \tag{34}$$

In equation (34), in the kth MCMC loop, $\theta'\theta$, $\theta' M\theta$, $\theta' M' M\theta$ and σ_u^2 are all scalars. Therefore, using the pre-calculated vector of $\log |I_J - \lambda M|$, we integrate $\log f(\lambda)$ over $(1/v^*_{min}, 1)$ to obtain the cumulative distribution of λ, and then the inverse sampling strategy can be operated.

When implementing this MCMC algorithm, the pre-calculated log Jacobian determinants, $\log |\boldsymbol{I}_N - \rho W|$ and $\log |I_J - \lambda M|$ are essential for saving running time when N and J are large. Many fast and accurate approximation methods for calculating the log Jacobian determinant have been proposed (Pace and Barry 1997; Barry and Pace 1999; Smirnov and Anselin 2001; Griffith 2004; LeSage and Pace 2009). A detailed summary and comparison of these methods are provided by Bivand, Hauke, and Kossowski (2013).

Application of the HSAR model

In this section, we demonstrate the HSAR model using a data set that consists of all leased residential land parcels from 2003 to 2009 in Beijing, China. There are 1,117 land parcels situated in 111 districts (*Jiedao*). A detailed data description is provided by Harris, Dong, and Zhang (2013). We regard land parcels as the lower level and districts as the regional level, thus giving a two-level hierarchical data structure.

Spatial interactions in house or land markets have been well recognized in at least two ways. The first regards the actual workings of the real estate market, in which the sale prices of nearby properties are often used to guide the price of a property for sale (e.g., Can 1992; Koschinsky, Lozano-Gracia, and Piras 2012). The second is related to the interacting decision-making process like house maintenance or repairing decisions among householders (Can and Megbolugbe 1997). It is also expected that regional effects are dependent. For example, regions of close spatial proximity might be similar in local tax rates, educational quality, public facility provision, and so on, which exert similar effects on land prices.

For model implementation, the lower level spatial weight matrix W is specified as

$$W_{ij} = exp\left(-\left(d_{ij}^2\right)/d^2, if\ d_{ij} \le d; 0, otherwise\right) \tag{35}$$

where d is the distance threshold, which is set to 1.5 km after a series of exploratory analyses on the empirical variograms of land prices (Cressie 1993; Banerjee, Carlin, and Gelfand 2004). The

Table 1 Estimation Results for the Land Price Data Using the HSAR and SAR models

	The HSAR model		The SAR model	
	Posterior mean	SE	Posterior mean	SE
Intercept	11.406*	0.988	10.314*	0.674
Logarea	–0.022	0.018	–0.020	0.018
LogDcbd	–0.380*	0.103	–0.234*	0.050
LogDsubway	–0.178*	0.041	–0.165*	0.034
LogDele	–0.011	0.040	–0.071*	0.031
LogDpark	–0.137*	0.061	–0.203*	0.045
LogDriver	0.097*	0.036	0.067*	0.029
Popden	0.019	0.013	0.023*	0.010
Buildings1949	–1.106*	0.515	–1.222*	0.293
Crimerate	0.001	0.008	0.003	0.004
Year dummies	Yes		Yes	
ρ	0.173	0.036	0.280	0.032
λ	0.829	0.124	NA	NA
σ_u^2	0.045	0.018	NA	NA
σ_e^2	0.575	0.026	0.633	0.027
DIC	3,036.7		3,235.6	
Pseudo R^2	0.423		0.311	

DIC, Deviance Information Criterion; NA, not applicable; SE, standard error; *, statistically significant at 95% credible level.

Euclidian distance between land parcels is d_{ij}. The regional-level spatial weight matrix M is based on the contiguity of the 111 districts. Both W and M are row normalized.

The outcome variable is the leasing price per square meter of each residential land parcel adjusted using the consumer price index (CPI). The lower level covariates used here are the proximity of each land parcel to the nearest subway stations (LogDsubway), elementary schools (LogDele), green parks (LogDpark), river (LogDriver) and the central business district (LogDcbd), and the land parcel size (Logarea). Year dummies (Year04–Year09) are included in the model to control the fixed period effects, with year 2003 as the baseline. The regional-level covariates used here are population density of each district (Popden), the proportion of houses built before 1949 (Buildings1949), and the number of violent crimes taking place per 1,000 people (Crimerate). Admittedly, variable selection is a problem—it is with a standard model and it is further complicated by a hierarchical one because we need to distinguish between the lower level covariates and the higher level ones.

The estimation results from the HSAR and classic SAR models are provided in Table 1. The inferences are based on three MCMC chains that each consist of 10,000 iterations with a burn-in period of 5,000 for both models.[2] Diffuse or quite non-informative priors are employed for model parameters. Two model fit indices are calculated for each model: the Pseudo R^2 and DIC with smaller values indicating a better model fit (LeSage et al. 2011). Both indices show that the HSAR model is better than the classic SAR in explaining the land price variations in Beijing.

From the regression coefficients estimates from the HSAR model in Table 1, we see that proximity to the CBD, subway stations, and parks has significant impacts on land prices. The

negative sign indicates that increasing accessibility to the CBD, subway stations, and parks will increase land prices. Proximity to rivers induces negative impacts to land prices, which probably mirrors the situation where most of the rivers in urban areas of Beijing are severely polluted specifically before the host of Olympic Games in 2008 (Harris, Dong, and Zhang 2013). As for the regional-level covariates (observed contextual effects), only the proportion of buildings built before 1949 (*Buildings1949*) is significant and exerts negative impacts on land prices.

As expected, we find spatial interactions at the land parcel level and dependent regional effects, indicated by the significant SAR parameters ρ (0.173) and λ (0.829) in the HSAR model. Hence, both the spatial dependence and group dependence are identified in our land price data set.

From Table 1, we see that there are not consistent patterns of differences between the two models with respect to regression coefficient estimates (β). However, we find that the introduction of a higher level impacts the precision estimate for each covariate. Generally, the variability in covariate effect estimates appears to increase in the HSAR model because of the inclusion of outcome variations at the higher level. We should be cautious about this observation though because the true process that generates the land prices is unknown. Nonetheless, we do find certain amount of variations at the district level (σ_u^2) that is not modeled in the classic SAR model. In a non-spatial hierarchical data structure, Goldstein (2003) has shown that ignoring group dependence will result in underestimated standard errors for covariate effects and the higher level covariates in particular. Differences between the HSAR model and the SAR model in terms of statistical inference on covariate effects are also observed. For example, the population density is found significant at the 95% confidence level in the SAR model whereas it is not so in the HSAR model, and the same situation is found for the effect of accesses to primary schools.

With respect to the estimates for ρ, it appears to be overestimated in the SAR model when compared with the HSAR model, and the 95% credible intervals estimates for ρ in the HSAR model do not contain the posterior mean of ρ from the SAR model. This indicates that at the land parcel level, spatial dependence is possibly confounded by the regional effects and thus incorrectly estimated as a result in the SAR model. More generally, if no effort is made to separate spatial interaction effects from the regional effects then we can anticipate that the two will be confounded.

As for the magnitude of covariate effects, the total, direct, and indirect impact summaries for each covariate and their 95% credible intervals from the HSAR and SAR models are provided in Table 2. For interpreting these measures, and taking the example of the effect of LogDsubway in the HSAR model, if the distance to subway stations decrease by 1%, land prices will increase by 0.178% from a direct effect, and by 0.037% from an indirect effect, producing a total effect of 0.216% increase.

The posterior means of regional (district) effects from HSAR are mapped in Fig. 3. The break points correspond to the lower quartile, median, and upper quartile of the regional effects. Quite as expected (as λ is significant and relatively large in magnitude), these regional effects exhibit a strong spatial pattern with high and low values clustering together, respectively.

Conclusions

In this article, we brought forward a research question: how to modify a standard spatial econometric method, specifically the SAR model, to accommodate a geographically hierarchical

Table 2 The Impact Estimates for Covariates with Significant Regression Coefficients Using Estimates from the HSAR and SAR, Respectively

	The HSAR model				The SAR model			
	Posterior mean	SE	2.5%	97.5%	Posterior mean	SE	2.5%	97.5%
Total impacts								
Intercept	13.816	1.313	11.292	16.452	14.354	1.137	12.230	16.664
LogDcbd	–0.459	0.125	–0.711	–0.214	–0.324	0.072	–0.467	–0.184
LogDsubway	–0.216	0.051	–0.317	–0.118	–0.232	0.049	–0.328	–0.136
LogDpark	–0.166	0.074	–0.312	–0.021	–0.284	0.063	–0.408	–0.161
LogDriver	0.118	0.043	0.037	0.205	0.094	0.040	0.018	0.174
Buildings1949	–1.341	0.630	–2.606	–0.127	–1.705	0.421	–2.532	–0.871
Direct impacts								
Intercept	11.446	0.991	9.483	13.361	10.413	0.675	9.090	11.715
LogDcbd	–0.381	0.103	–0.585	–0.178	–0.235	0.051	–0.334	–0.135
LogDsubway	–0.178	0.041	–0.258	–0.098	–0.168	0.035	–0.236	–0.100
LogDpark	–0.137	0.061	–0.257	–0.017	–0.206	0.045	–0.294	–0.119
LogDriver	0.098	0.036	0.031	0.169	0.068	0.029	0.013	0.125
Buildings1949	–1.110	0.517	–2.145	–0.106	–1.237	0.300	–1.817	–0.629
Indirect impacts								
Intercept	2.370	0.618	1.240	3.671	3.941	0.674	2.716	5.367
LogDcbd	–0.079	0.028	–0.142	–0.030	–0.089	0.024	–0.140	–0.047
LogDsubway	–0.037	0.013	–0.065	–0.015	–0.064	0.017	–0.100	–0.034
LogDpark	–0.029	0.015	–0.063	–0.003	–0.078	0.021	–0.123	–0.040
LogDriver	0.020	0.009	0.006	0.040	0.026	0.012	0.005	0.051
Buildings1949	–0.231	0.127	–0.519	–0.019	–0.468	0.137	–0.761	–0.224

SE, standard error.

data structure? To answer this question, we began by outlining a typology of four different model specifications that are distinguished by the type of weights matrix used and whether the regional effects are treated as fixed or random. Previous research and implementations of these models were discussed.

In this article, we have given extensive attention to the HSAR model because of its methodological advantage that it allows for estimation of spatial spillover effects while also allowing for the control and examination of dependent group effects. By simultaneously modeling the spatial dependence and the group dependence, the HSAR model avoids the situation where these two types of dependence are confounded. It provides more accurate estimates for both of them. Our view is that the HSAR is especially well suited to the increasingly available hierarchical data sets with detailed geographic information and for providing a greater synthesis between multilevel modeling and spatial econometric perspectives (more so than the other model specifications).

We end this article by illustrating several further elaborations on the HSAR. First, other standard spatial econometric models like spatial error model and spatial Durbin model could be easily extended in this proposed framework as they have very similar likelihood functions as a

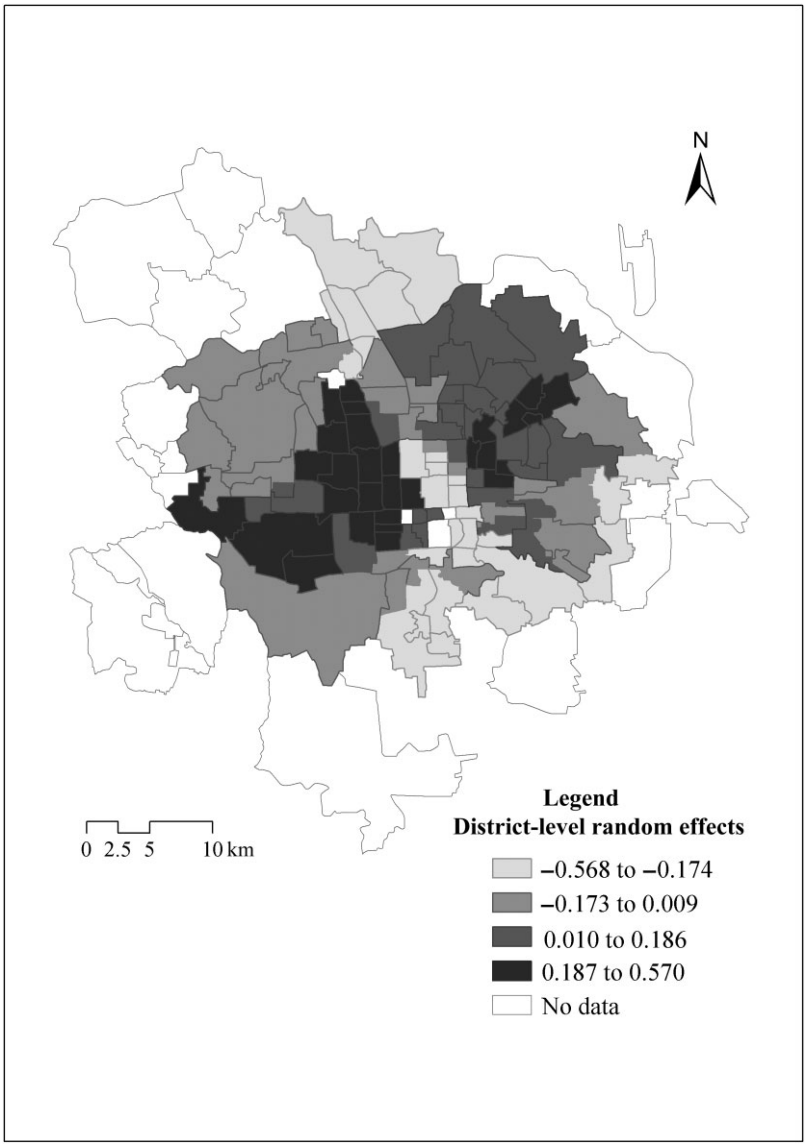

Figure 3. The district (regional) level random effects from the hierarchical spatial autoregressive model.

SAR model. In essence, HSAR just allows the model intercept term to have a spatial dependence structure through the simultaneous autoregressive process of θ. The future extension could be to allow for differentiating spatial processes for the regression coefficients, which enables us to simultaneously explore spatial heterogeneity. However, the difficulty for this lies in two aspects. The first concerns whether a univariate SAR could be extended to a proper multivariate SAR. The second concerns model estimation. By assigning varying SAR processes to regression coefficients, the autoregressive parameters may not be well identified even in the Bayesian framework because of the possible strong correlations between them and of the model complexity.

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Notes

1 We are grateful to a referee for identifying this property of the model.

2 The MCMC sampler for HSAR is coded using the R language. The convergence of the MCMC sampler for HSAR is diagnosed using the CODA package (Plummer et al. 2006) in R. Because of the use of sparse matrices for calculating the determinants of W and M and also the use of inversion sampling for updating SAR parameters ρ and λ, the numerical computation is not especially time consuming. It takes about 3 min for the MCMC sampler to produce 10,000 draws on a laptop with a 2.5 GHz processor. Before applying the code to the land price data, we conduct a series of simulation studies with known data-generating processes and using the geography of the land parcel data to test the code, the results show that the HSAR model could accurately retrieve the true model parameters. Both the R code for implementing the HSAR model and the simulation study results are available upon request.

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